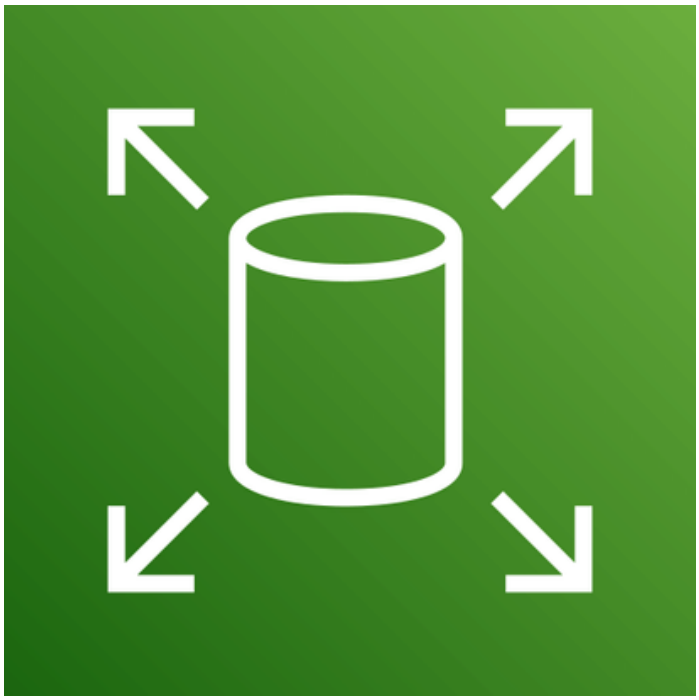


AWS EBS

Volume

Snapshots

Lifecycle Manager

Concept Overview:

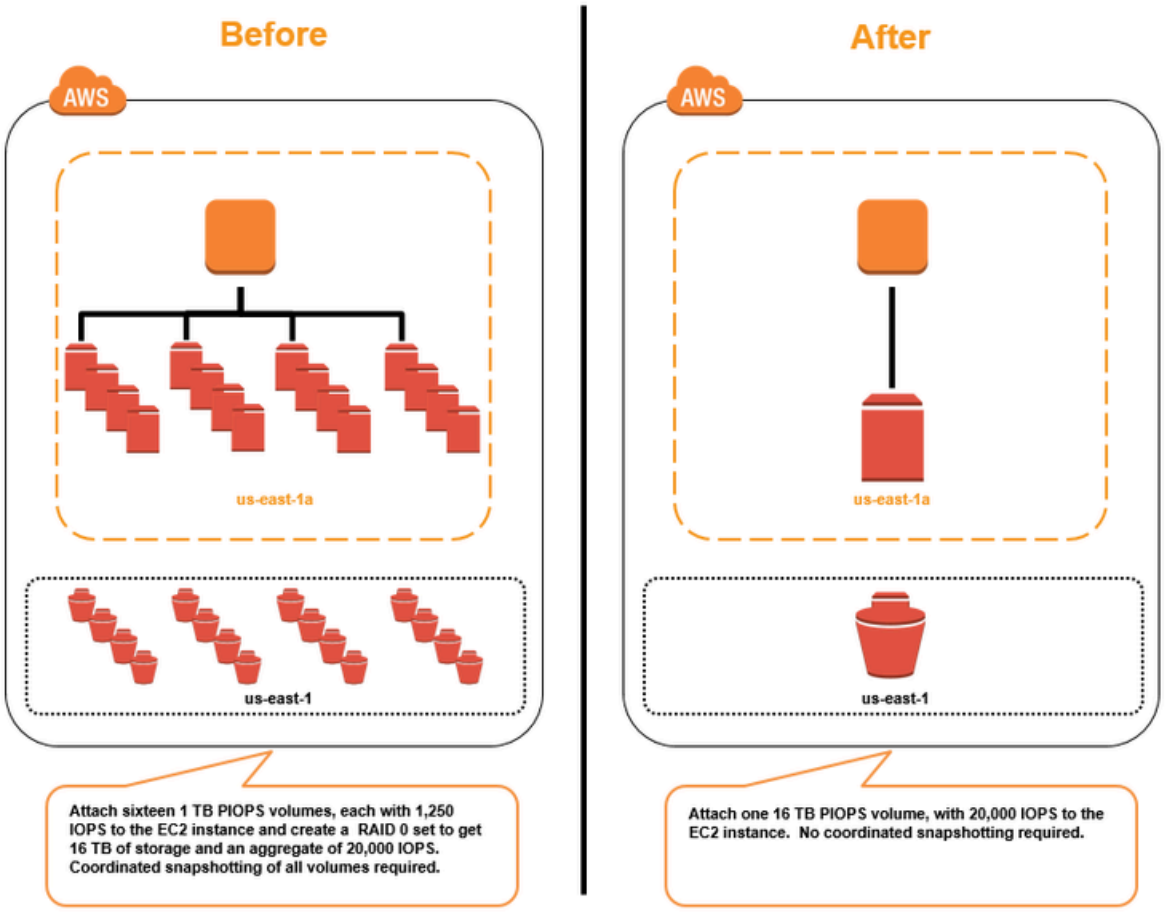
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Advantages of EBS	2
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About EBS:

EBS(Elastic Block Store) is a block-type type durable and persistent storage that can be attached to EC2 instances for additional storage.

It allows users to create storage volumes and attach them to EC2 instances, acting like a hard drive for those instances.

Each EC2 instance is a virtual server, and its instance type decides the amount of CPU, memory, storage, and networking it gets.



Advantages of EBS:

- Multiple volume types: Amazon EBS provides multiple volume types that allow you to optimize storage performance and cost for a broad range of applications. Volume types are divided into two major categories: SSD-backed storage for transactional workloads, and HDD-backed storage for throughput intensive workloads.

- Scalability: You can create Amazon EBS volumes with capacity and performance specifications that meet your needs. As your needs changes, you can use Elastic Volumes operations to dynamically increase capacity or tune performance, with no downtime.

- Backup and recovery: Use Amazon EBS snapshots to back up the data stored on your volumes. You can then use those snapshots to instantly restore volumes or to migrate data across AWS accounts, AWS Regions, or Availability Zones.

- Data protection: Use Amazon EBS encryption to encrypt your Amazon EBS volumes and Amazon EBS snapshots. Encryption operations occur on the servers that host Amazon EC2 instances, ensuring the security of both data-at-rest and data-in-transit between an instance and its attached volume and subsequent snapshots.

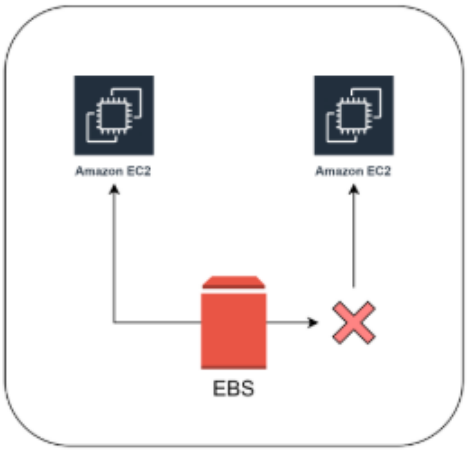
Advantages of EBS:

- Data availability and durability: io2 Block Express volumes provide 99.999% durability with an annual failure rate of 0.001%. Other volume types provide 99.8% to 99.9% durability with an annual failure rate of 0.1% to 0.2%. Additionally, volume data is automatically replicated across multiple servers in an Availability Zone to prevent the loss of data from the failure of any single component.

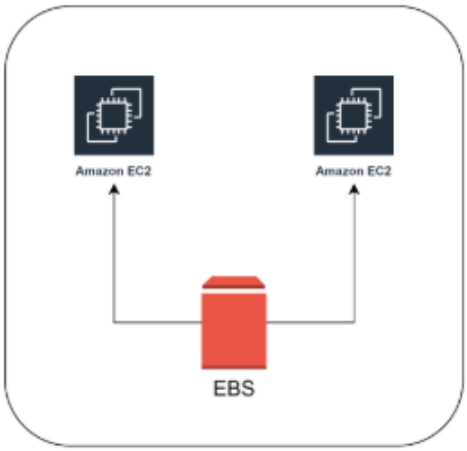
- Data archiving: EBS Snapshots Archive provides a low-cost storage tier to archive full, point-in-time copies of EBS Snapshots that you must retain for 90 days or more for regulatory and compliance reasons, or for future project releases.

One more thing, one EBS can't connect with multiple EC2 by default without EBS Multi-Attach support.

Default



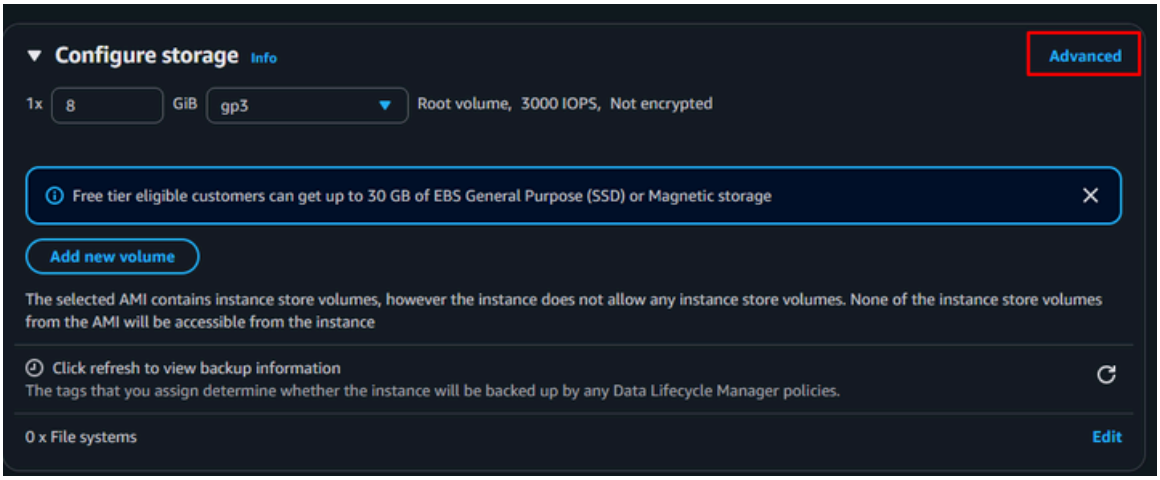
With Multi-Attach



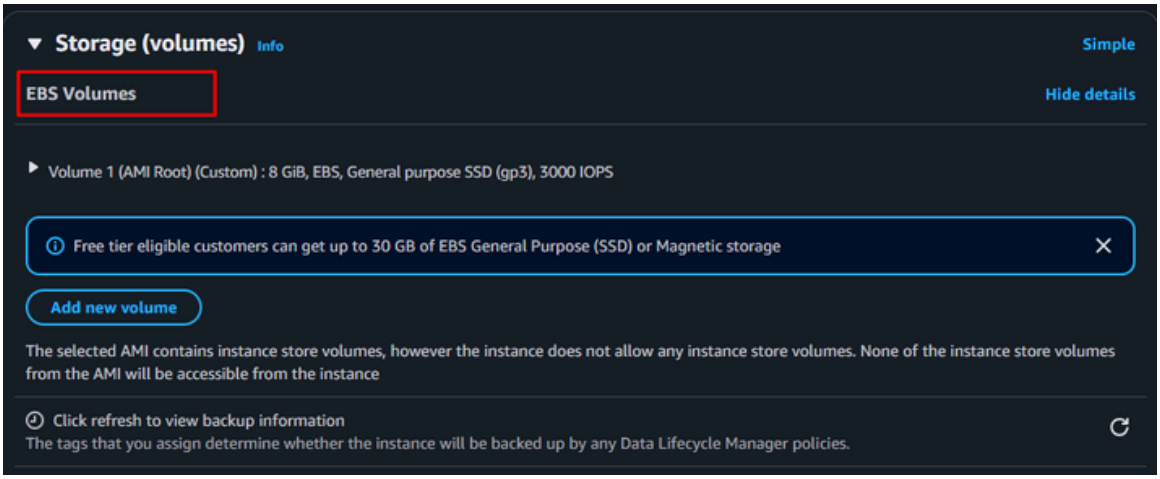
Create EBS:

Before creating EBS let’s launch a EC2 instace first.

To launch a new instance, log in to your AWS account and search for EC2. Then, name and configure your instance, and launch it. If you want, you can add EBS during your EC2 creation process, which is located in the Storage configuration section.

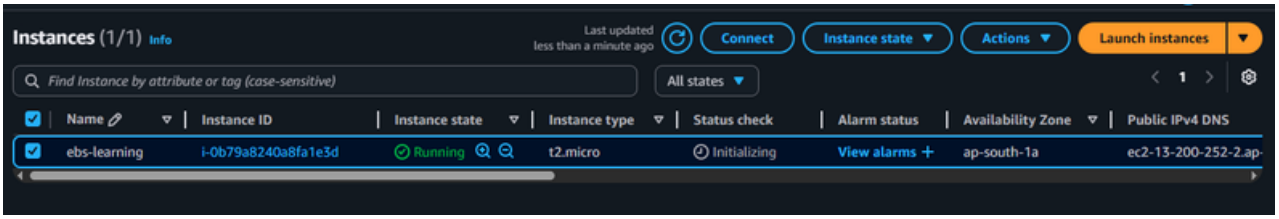


Click on Advance and get the EBS volume.

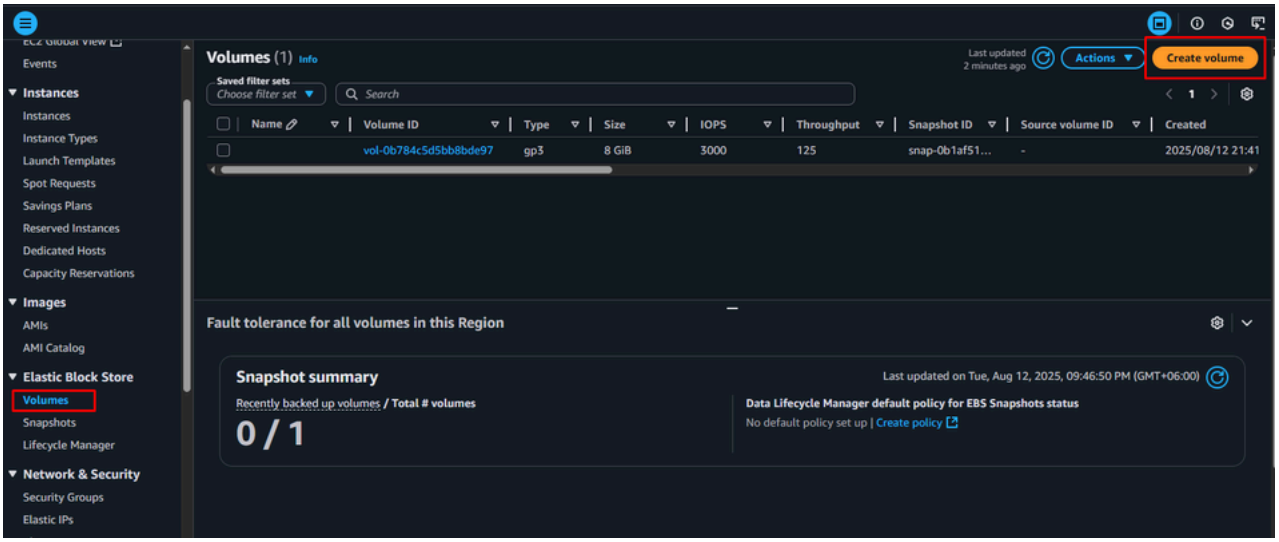


Create EBS:

In my case, I kept everything as it is. Here is my instance, let’s connect to it and then create an EBS volume.

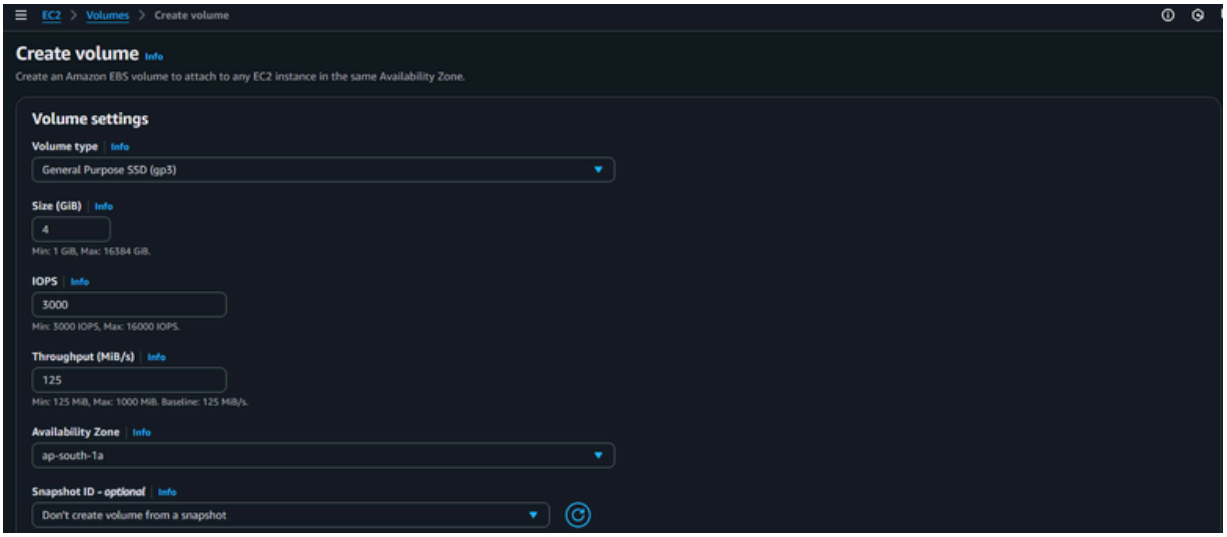


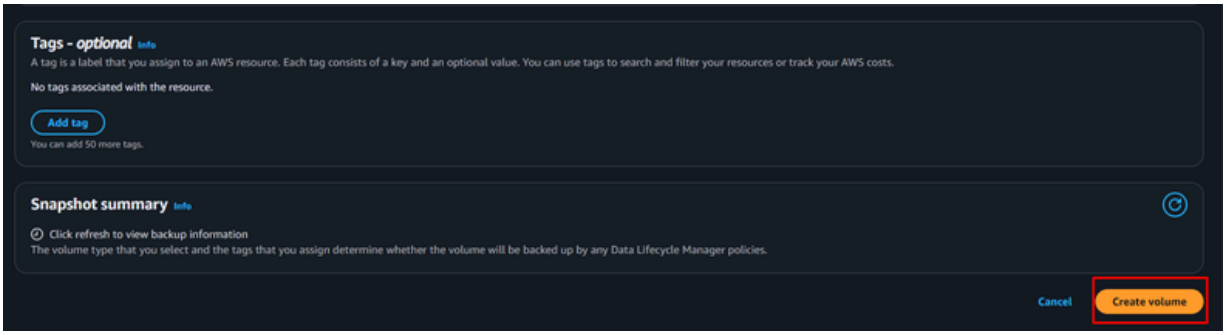
It’s time to create a new EBS volume. Let’s start: first, go to the Volumes menu under the EBS section, and then click the “Create Volume” button in the top-right corner. There is already one volume available, which was created when the EC2 instance was launched.



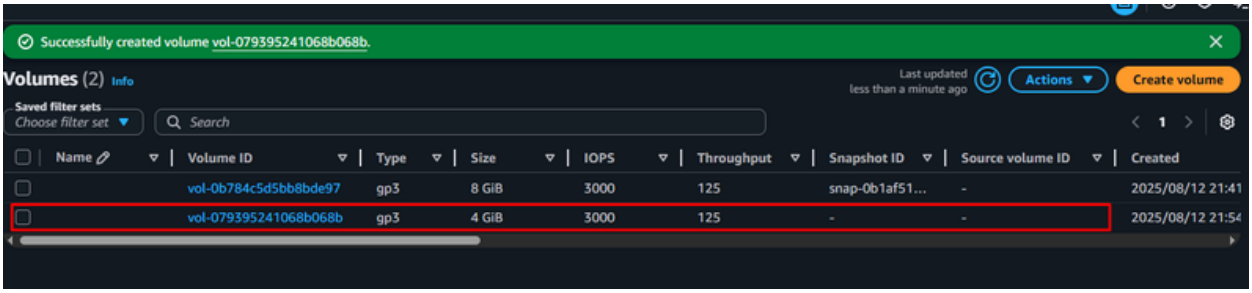
Create EBS:

Configure the volume based on your need, in my case i kept it minimal.



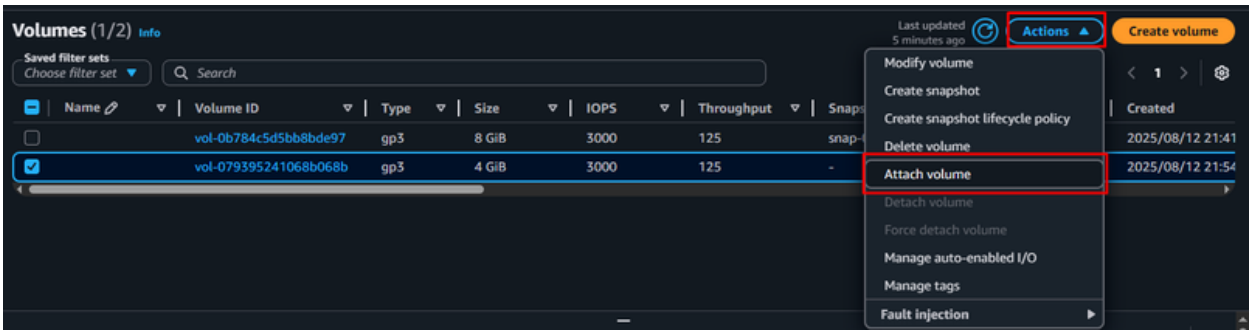


It’s created successfully.

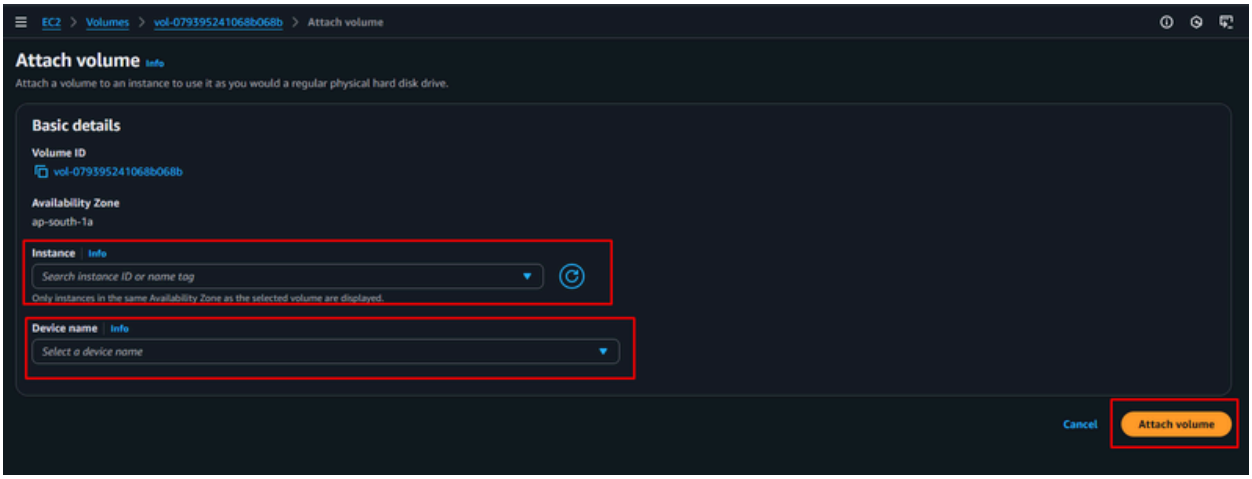


Attaching New Volume:

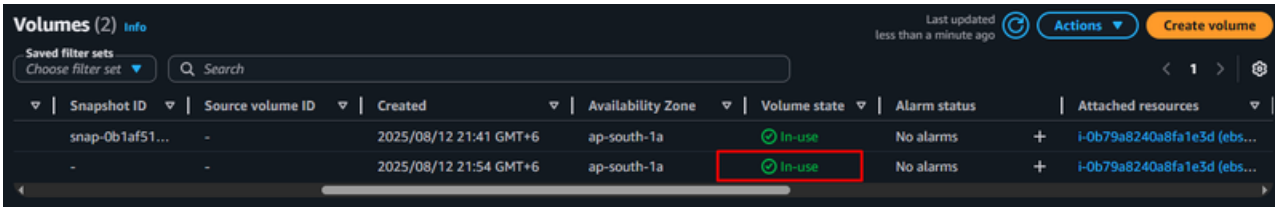
Now attach our newly created volume. For attaching volume first you need to select this volumne and click on action button and then click on attach volume button.



Then you need to select the instance and provide the Device name, then click on attach volume button.



Now check it properly works or not.



Attaching New Volume:

Check the list block devices before attaching volume and after.

Before → default instance

```
ubuntu@ip-172-31-32-51:~$ lsblk
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0 27.2M  1 loop /snap/amazon-ssm-agent/11320
loop1       7:1    0 73.9M  1 loop /snap/core22/1981
loop2       7:2    0 50.9M  1 loop /snap/snapd/24505
xvda      202:0    0    8G  0 disk
├─xvda1   202:1    0    7G  0 part /
├─xvda14  202:14   0    4M  0 part
├─xvda15  202:15   0  106M  0 part /boot/efi
└─xvda16  259:0    0  913M  0 part /boot
ubuntu@ip-172-31-32-51:~$
```

After → Attach Volume

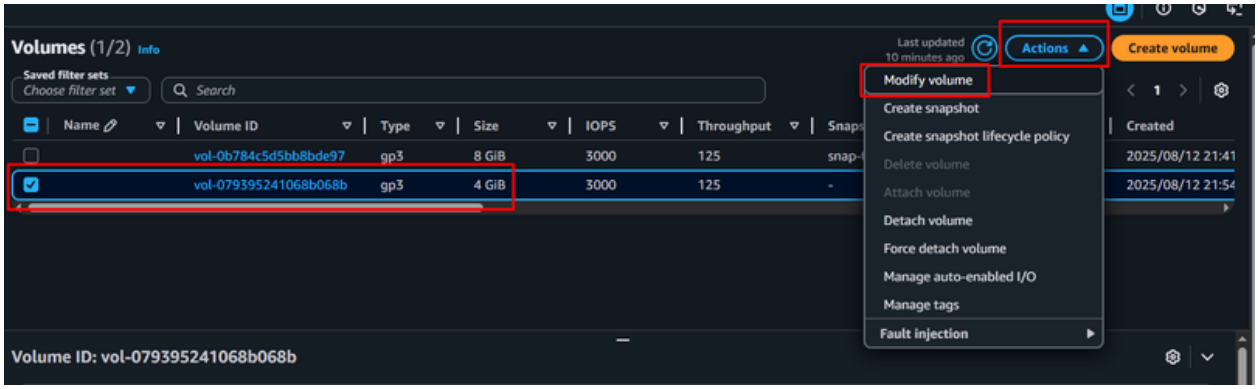
```
ubuntu@ip-172-31-32-51:~$ lsblk
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0 27.2M  1 loop /snap/amazon-ssm-agent/11320
loop1       7:1    0 73.9M  1 loop /snap/core22/1981
loop2       7:2    0 50.9M  1 loop /snap/snapd/24505
xvda      202:0    0    8G  0 disk
├─xvda1   202:1    0    7G  0 part /
├─xvda14  202:14   0    4M  0 part
├─xvda15  202:15   0  106M  0 part /boot/efi
└─xvda16  259:0    0  913M  0 part /boot
xvdh      202:112  0    4G  0 disk
ubuntu@ip-172-31-32-51:~$
```

Great, it attaching successfully with our EC2 instance.

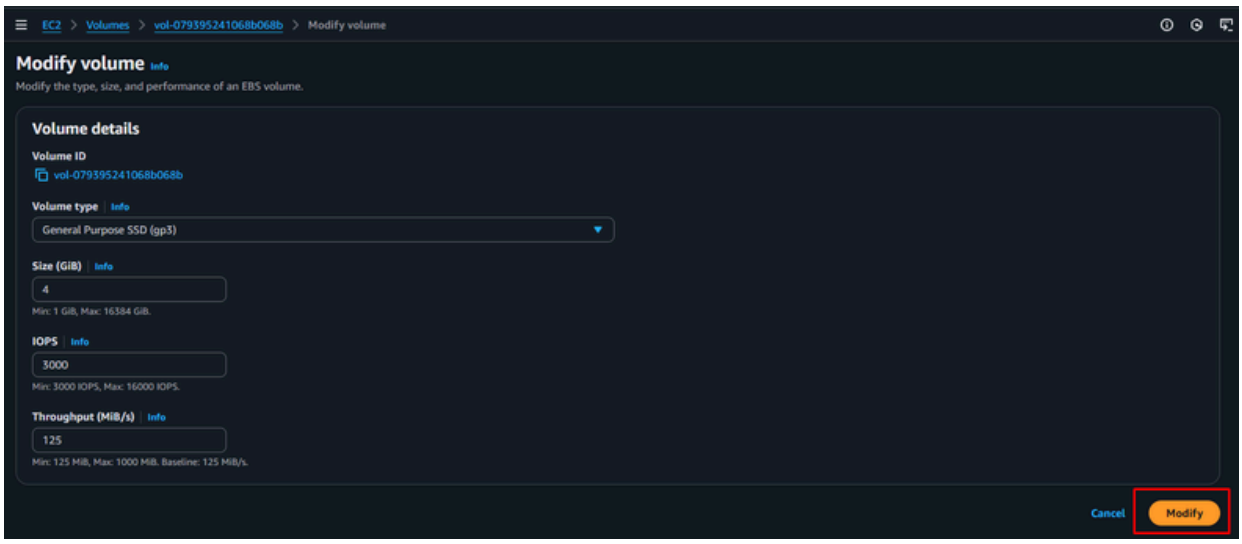
Modify EBS:

If you need to modify your existing EBS, you can do so. Let’s look at an example.

First, select your volume, then click the “Actions” button in the top-right corner, and choose “Modify Volume.”



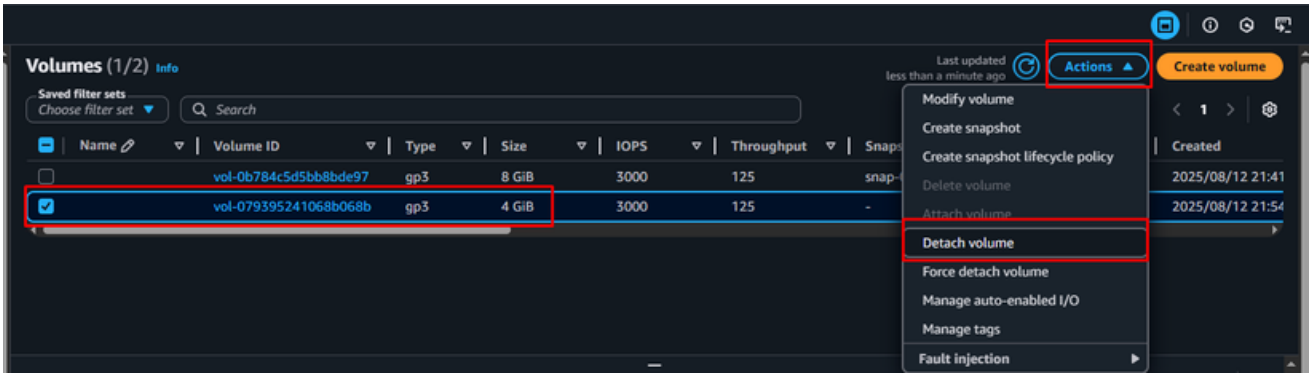
Then get this page. Modify volume based on your need and click on modify button for save it.



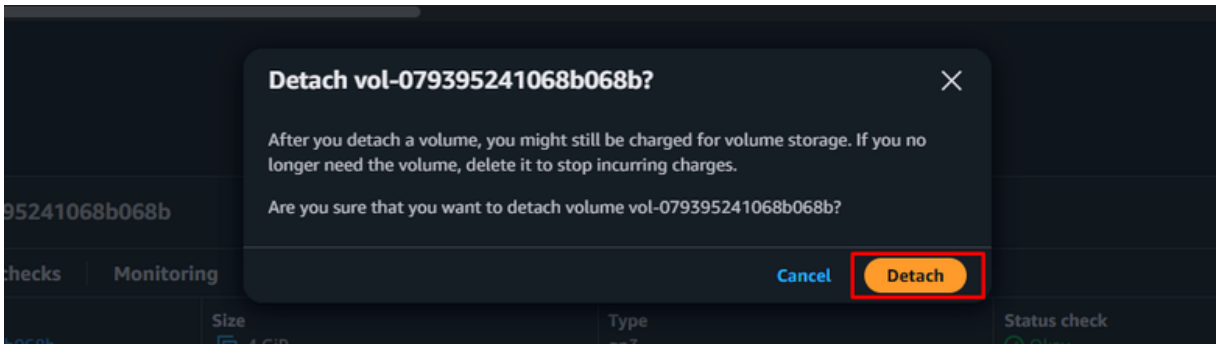
Detach & Delete EBS:

If you don’t need this volume, you can detach or delete it directly. Let’s see how you can do this.

First, select your volume, then click the “Actions” button in the top-right corner, and choose “Detach Volume.”



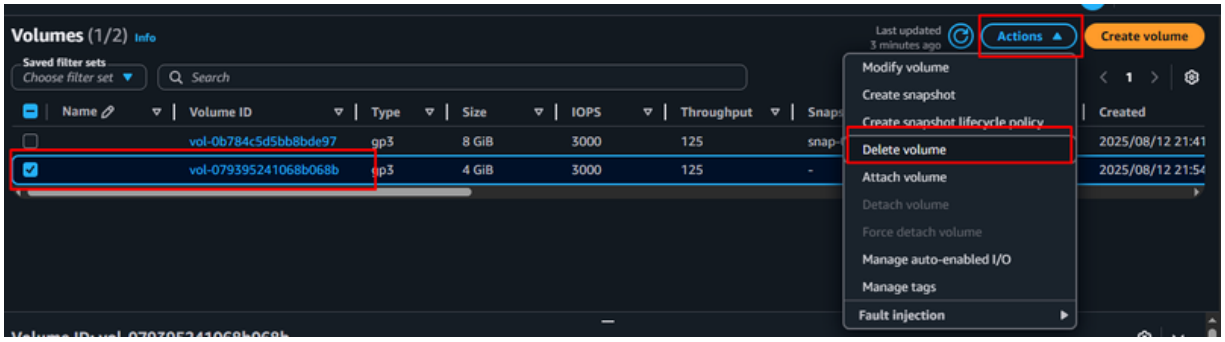
Then get this popup for confirming you.



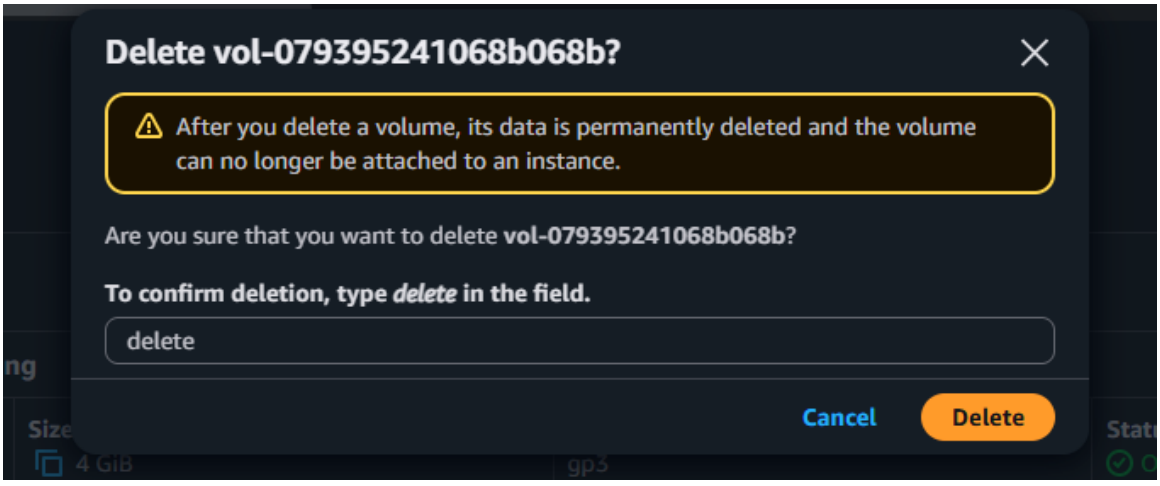
Detach & Delete EBS:

If you don’t need this volume you delete it.

First, select your volume, then click the “Actions” button in the top-right corner, and choose “Delete Volume.”



Then get this popup for confirming you. Type delete on input field and click delete button. I will be delete.

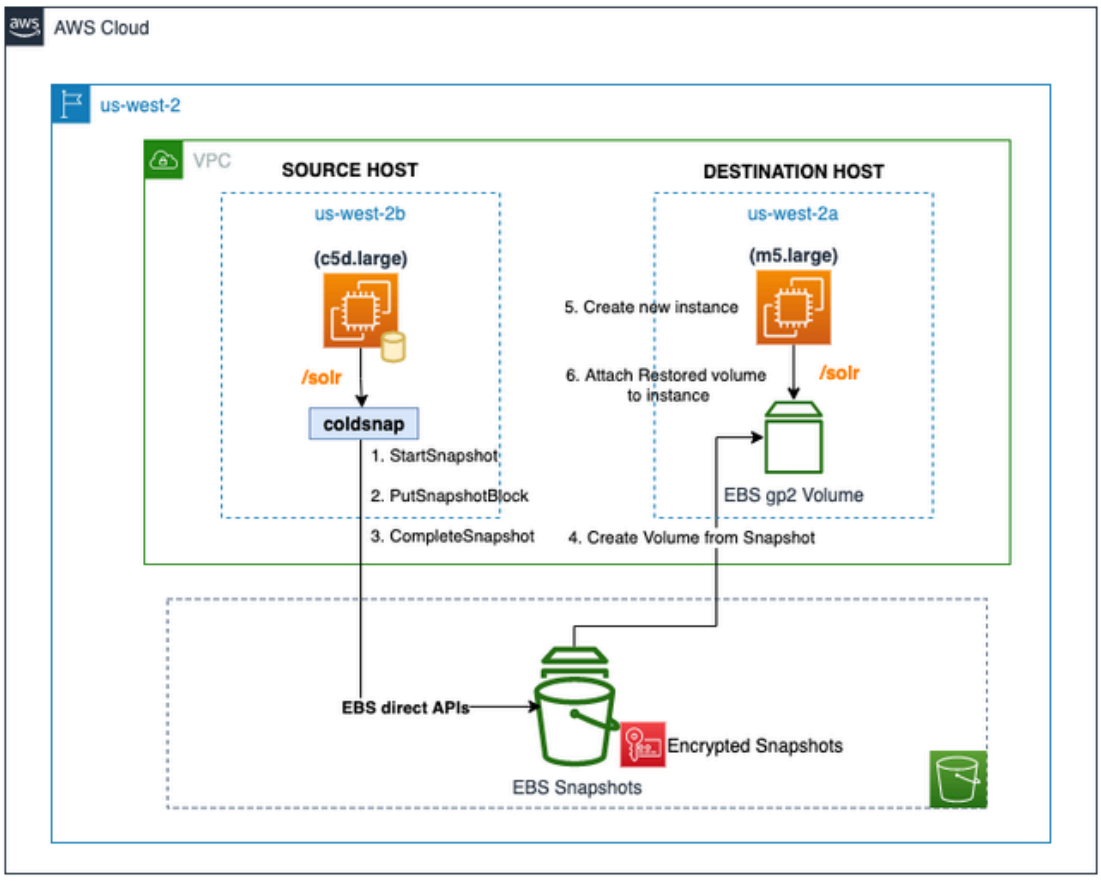


Basic understanding of EBS Snapshots:

You can back up the data on your Amazon EBS volumes by making point-in-time copies, known as Amazon EBS snapshots. A snapshot is an incremental backup, which means that we save only the blocks on the volume that have changed since the most recent snapshot. This minimizes the time required to create the snapshot and saves on storage costs by not duplicating data.

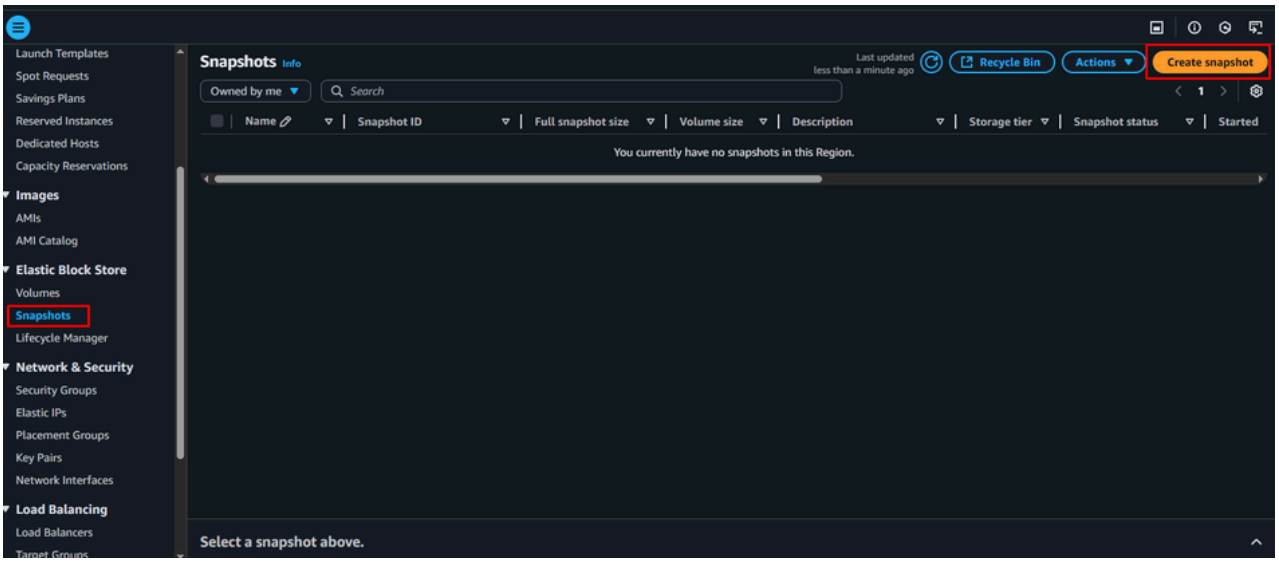
EBS Snapshots are mainly used to back up your volume and protect your data from accidental issues.

Here the diagram for better understanding.

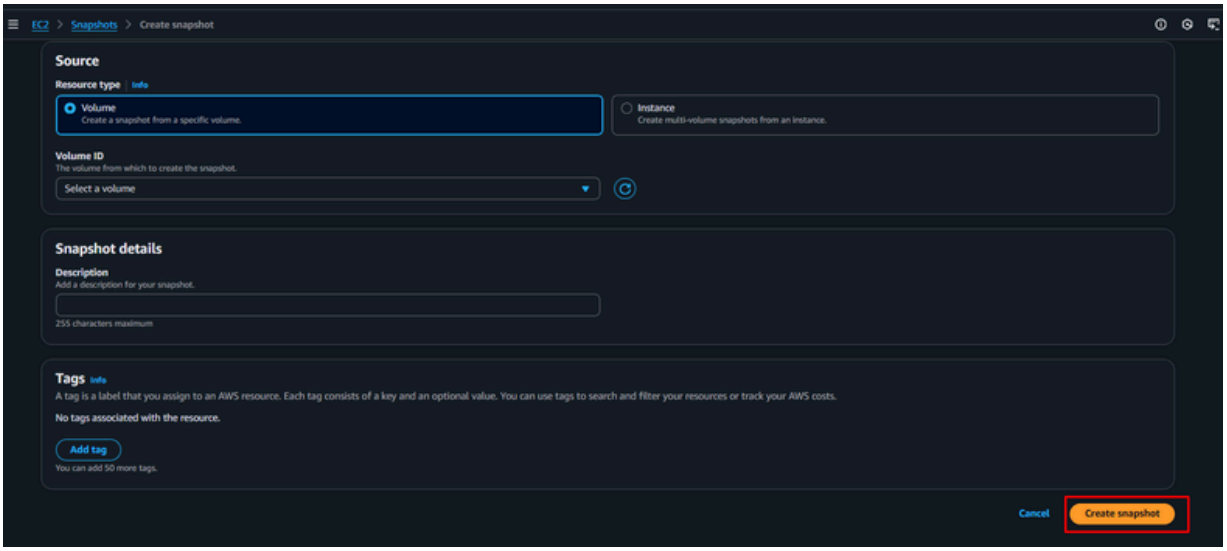


Basic understanding of EBS Snapshots:

You can access it from EBS menu snapshots section.



Configure it based on your need and create.



Basic understanding of EBS Lifecycle Manager:

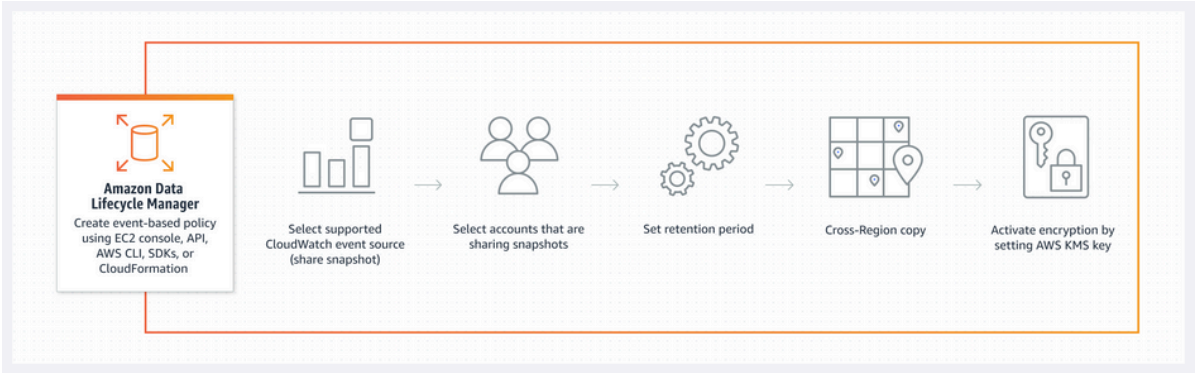
a service that automates the management of Amazon Elastic Block Storage (EBS) snapshots and EBS-backed Amazon Machine Images (AMIs).

It allows users to create policies that automate the creation, retention, and deletion of these backups, streamlining data protection and reducing storage costs, according to AWS.

It help you to:

- Backup automation: Create snapshots daily, weekly, or at any custom interval.
- Retention management: Automatically delete older snapshots to save storage costs.
- Consistency: Ensure all critical EBS volumes are regularly backed up.

Follow this figure for better understandig:

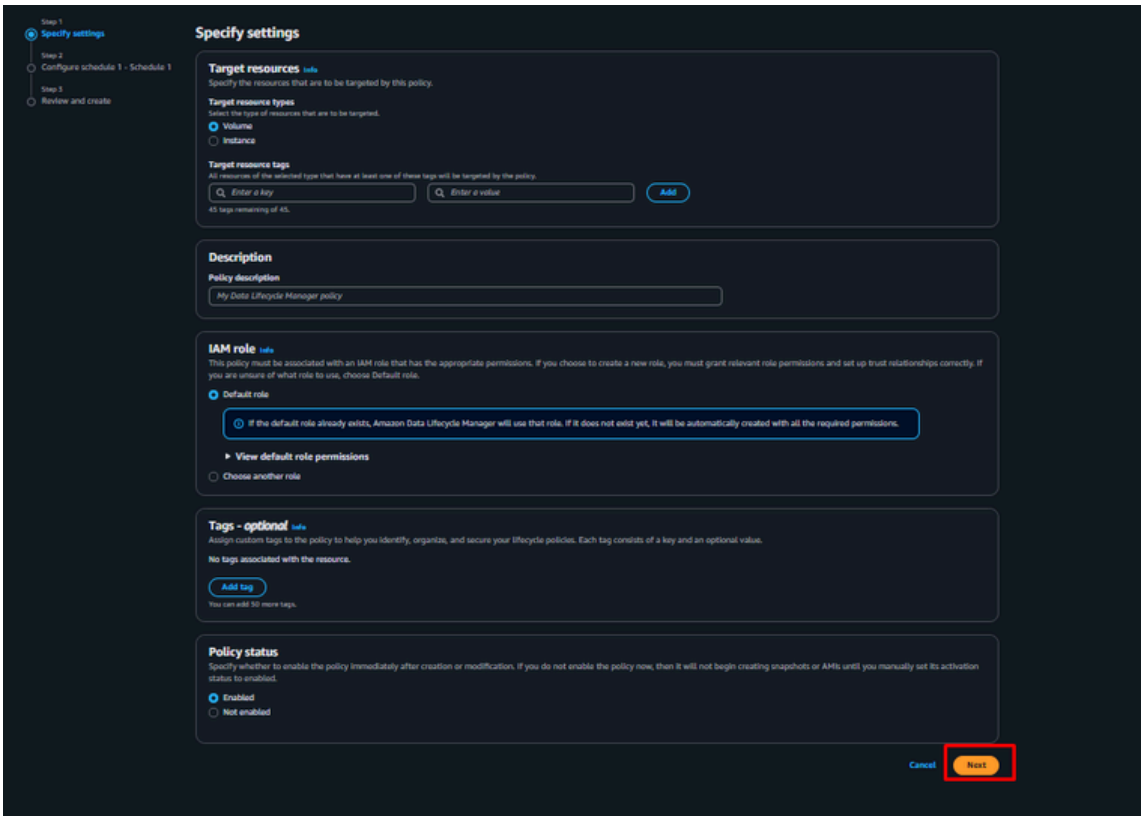


Basic understanding of EBS Lifecycle Manager:

You can access it from EBS menu lifecycle manager section.



Configure it based on your need step by step like this.



Thank You

Stay Connect:

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/alamgirweb11